When Concurrency Matters: Behaviour-Oriented Concurrency

LUKE CHEESEMAN, Imperial College London, UK
MATTHEW J. PARKINSON, Microsoft Azure Research, UK
SYLVAN CLEBSCH, Microsoft Azure Research, UK
MARIOS KOGIAS, Imperial College London, UK Microsoft Research, UK
SOPHIA DROSSOPOULOU, Imperial College London, UK
DAVID CHISNALL, Microsoft, UK
TOBIAS WRIGSTAD, Uppsala University, Sweden
PAUL LIÉTAR, Imperial College London, UK

We have a semantics and implementations

$$\textsc{Step}\ \frac{E, h \hookrightarrow E', h'}{R \uplus (\overline{\kappa}, E), P, h \rightsquigarrow R \uplus (\overline{\kappa}, E'), P, h'}$$

$$\textsc{Spawn}\ \frac{E \hookrightarrow_{\texttt{when}\ (\overline{\kappa'})\ \{E''\}} E'}{R \uplus (\overline{\kappa}, E), P, h \rightsquigarrow R \uplus (\overline{\kappa}, E'), P : (\overline{\kappa'}, E''), h}$$

$$\textsc{Run}\ \frac{\left(\bigcup_{(\overline{\kappa'},\_) \in (P' \cup R)} \overline{\kappa'}\right) \cap \overline{\kappa} = \varnothing}{R, P' : (\overline{\kappa}, E) : P'', h \rightsquigarrow R \uplus (\overline{\kappa}, E), P' : P'', h}$$

$$\textsc{End}\ \frac{\mathit{finished}(E)}{R \uplus (\overline{\kappa}, E), P, h \rightsquigarrow R, P, h}$$

Behavior-Oriented Concurrency in Python

Lock-less, Deadlock-free, Ownership-based https://microsoft.github.io/bocpy/

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$$\text{STEP}\ \frac{E, h \hookrightarrow E', h'}{R \uplus (\overline{\kappa}, E), P, h \rightsquigarrow R \uplus (\overline{\kappa}, E'), P, h'} \qquad \text{SPAWN}\ \frac{E \hookrightarrow_{\text{when } (\overline{\kappa'})\,\{E''\}} E'}{R \uplus (\overline{\kappa}, E), P, h \rightsquigarrow R \uplus (\overline{\kappa}, E'), P : (\overline{\kappa'}, E''), h}$$

$$\text{RUN}\ \frac{\left(\bigcup_{(\overline{\kappa'},\_)\in(P' \cup R)} \overline{\kappa'}\right) \cap \overline{\kappa} = \varnothing}{R, P' : (\overline{\kappa}, E) : P'', h \rightsquigarrow R \uplus (\overline{\kappa}, E), P' : P'', h} \qquad \text{END}\ \frac{\mathit{finished}(E)}{R \uplus (\overline{\kappa}, E), P, h \rightsquigarrow R, P, h}$$

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The semantics have incidental restrictions on executions